

Question Paper Preview

Notations :

- 1.Options shown in **green** color and with  icon are correct.
- 2.Options shown in **red** color and with  icon are incorrect.

Change Font Color :	No
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Help Button :	No
Show Reports :	No
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MLT

Section Id :	640653124271
Number of Questions :	15
Section Marks :	50
Maximum Instruction Time :	0

Question Number : 1 Question Id : 6406531852220 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : MACHINE LEARNING TECHNIQUES (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406536004251.  Yes

6406536004252.  No

Question Number : 2 Question Id : 6406531852221 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Consider a dataset of 100 points in $\mathbb{R}^5$. The eigenvalues of the sample covariance matrix are:
4.50, 2.00, 0.80, 0.05, 0.01.

If we run PCA on this dataset and retain the top- k principal components, what is a good choice of k using the heuristic that we retain the smallest k such that the retained components explain at least 95% of the total variance?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

3

Question Number : 3 Question Id : 6406531852222 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Using the polynomial kernel

$$K(f, g) = (f^T g + 1)^2,$$

what is the dimension of the transformed feature space when the original input vector has dimension 3?

Options :

6406536004254. ✖ 6

6406536004255. ✖ 7

6406536004256. ✖ 9

6406536004257. ✔ 10

Question Number : 4 Question Id : 6406531852223 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Suppose you toss a coin 100 times and observe $n = 47$ heads. Let θ denote the probability of heads on each flip. Assume the following discrete prior over θ :

$$p(\theta) = \begin{cases} 0.4 & \theta = 0.5, \\ 0.6 & \theta = 0.4, \\ 0 & \text{otherwise} \end{cases}.$$

What will be the maximum a posteriori estimate (MAP) of the parameter θ ?

Options :

6406536004258. ✖ 0.4

6406536004259. ✔ 0.5

6406536004260. ✖ 0.45

6406536004261. ✖ 0.6

Question Number : 5 Question Id : 6406531852224 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Consider a linear regression model $\hat{y} = X^T w$, where $X \in \mathbb{R}^{d \times n}$, $y \in \mathbb{R}^{n \times 1}$, and $w \in \mathbb{R}^{d \times 1}$. The squared-error loss function is defined as:

$$L(w) = \frac{1}{n} \|\hat{y} - y\|_2^2 = \frac{1}{n} \|X^T w - y\|_2^2,$$

where n is the number of samples. Which of the following equations correctly represents the gradient $\frac{\partial L}{\partial w}$?

Options :

6406536004262. ✖ $\frac{\partial L}{\partial w} = \frac{1}{n}(\hat{y} - y)X^T$

6406536004263. ✖ $\frac{\partial L}{\partial w} = \frac{1}{n}X(\hat{y} - y)$

6406536004264. ✖ $\frac{\partial L}{\partial w} = \frac{2}{n}(\hat{y} - y)X^T$

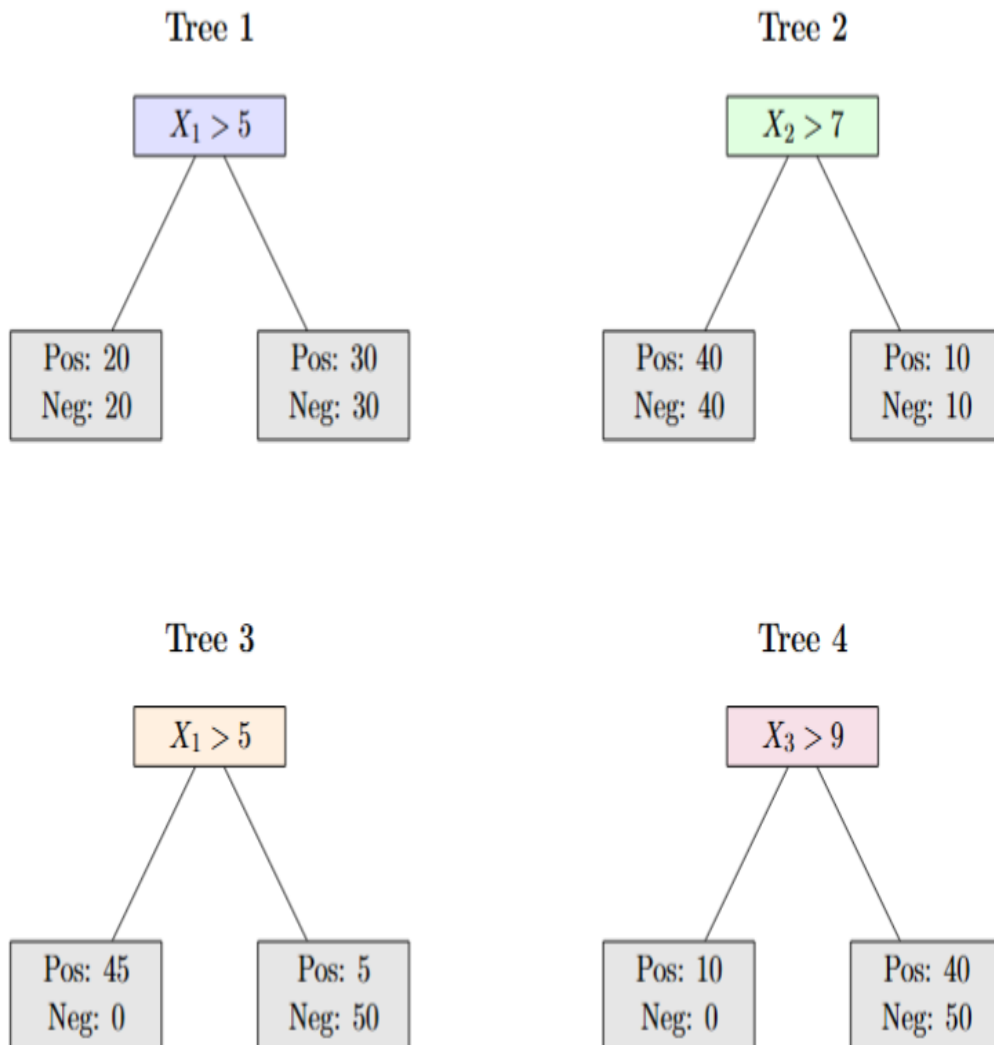
6406536004265. ✔ $\frac{\partial L}{\partial w} = \frac{2}{n}X(\hat{y} - y)$

Question Number : 6 Question Id : 6406531852225 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Suppose you have a binary classification problem with 3-dimensional feature vectors $\mathbf{x} \in \mathbb{R}^3$. You are given 50 positive and 50 negative training examples, and you want to build a decision tree classifier. Consider 4 possible splits at the root node:



Which of the following splits gives the highest information gain?

Options :

6406536004266. ✖ Tree 1

6406536004267. ✖ Tree 2

6406536004268. ✔ Tree 3

6406536004269. ✖ Tree 4

Question Number : 7 Question Id : 6406531852226 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Suppose we want to learn $P(y \mid x_1, x_2)$ using the Naive Bayes classifier, where $y \in \{0, 1\}$ is a binary label, and $x_1, x_2 \in \{0, 1\}$ are binary features. Then, which of the following will be true?

Options :

6406536004270. ✖ Gaussian Naive Bayes would be a good choice to model $P(x_i \mid y)$.

6406536004271. ✔ Bernoulli Naive Bayes would be a good choice to model $P(x_i \mid y)$.

6406536004272. ✔ The number of parameters to estimate is 5.

6406536004273. ✖ The number of parameters to estimate is 3.

Question Number : 8 Question Id : 6406531852227 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

In logistic regression, the batch gradient descent update rule is

$$w_{t+1} = w_t - \eta \cdot \frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i) x_i,$$

where

$$\hat{y}_i = \sigma(w_t^\top x_i).$$

For the current mini-batch containing two training samples

$$x_1 = \begin{bmatrix} 2 \\ -1 \end{bmatrix}, \quad y_1 = 1, \quad x_2 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \quad y_2 = 0,$$

and the current weight vector and learning rate

$$w_t = \begin{bmatrix} 0.5 \\ 0.2 \end{bmatrix}, \quad \eta = 0.1,$$

compute the updated weight vector w_{t+1} .

Options :

6406536004274. ✖ $\begin{bmatrix} 0.4935 \\ 0.0720 \end{bmatrix}$

6406536004275. ✖ $\begin{bmatrix} 0.5102 \\ 0.1840 \end{bmatrix}$

6406536004276. ✔ $\begin{bmatrix} 0.4628 \\ 0.0984 \end{bmatrix}$

6406536004277. ✖ $\begin{bmatrix} 0.5481 \\ 0.0123 \end{bmatrix}$

Question Number : 9 Question Id : 6406531852228 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Consider a linearly separable dataset $D = \{(x_i, y_i)\}_{i=1}^n$, where each feature vector satisfies $\|x_i\| \leq 1$. Suppose there exists a weight vector w^* that correctly classifies all examples with margin $\gamma = \frac{1}{2}$. Let the perceptron algorithm be run on this dataset and produce a separating weight vector w_{perc} , with margin is $\frac{1}{3}$. What will be the upper bound on the number of mistakes the perceptron can make?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

4

Question Id : 6406531852229 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (10 to 11)

Question Label : Comprehension

Consider the transformation

$$\phi : \mathbb{R}^2 \rightarrow \mathbb{R}^6$$

associated with the polynomial kernel of degree (2):

$$\phi(x) = \begin{bmatrix} 1 \\ x_1 \\ x_2 \\ x_1^2 \\ x_2^2 \\ 2x_1x_2 \end{bmatrix}.$$

A kernel-SVM is trained using this kernel, and the optimal weight vector is:

$$w^* = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}.$$

You may assume the dataset is linearly separable in the transformed space.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 10 Question Id : 6406531852230 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

What is the shape of the decision boundary in $\mathbb{R}^2$?

Options :

6406536004279. ✓ It is a parabola of the form $x_2 = -x_1^2$

6406536004280. ✗ It is a hyperbola

6406536004281. ✗ It is a straight line

6406536004282. ✗ It is a circle

Question Number : 11 Question Id : 6406531852231 Question Type : MSQ

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following training data-points are certainly **not** support vectors?

Options :

6406536004283. ✖ $\begin{bmatrix} 0 \\ -0.5 \end{bmatrix}$

6406536004284. ✖ $\begin{bmatrix} 1 \\ -1.2 \end{bmatrix}$

6406536004285. ✔ $\begin{bmatrix} 2 \\ -6 \end{bmatrix}$

6406536004286. ✔ $\begin{bmatrix} -1 \\ -3 \end{bmatrix}$

6406536004287. ✔ $\begin{bmatrix} 2 \\ 4 \end{bmatrix}$

Question Number : 12 Question Id : 6406531852232 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

For linearly separable data, a hard-margin SVM minimizes $\|w\|^2$ subject to the constraints $(w^T x_i)y_i \geq 1$ for all i . If one of the training samples is removed, what may happen to the decision boundary?

Options :

6406536004288. ✖ It always remains unchanged.

6406536004289. ✔ It may shift toward the removed data point.

6406536004290. ✖ It shifts away from the removed data point in all cases.

6406536004291. ✖ It becomes non-linear.

Question Id : 6406531852233 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (13 to 14)

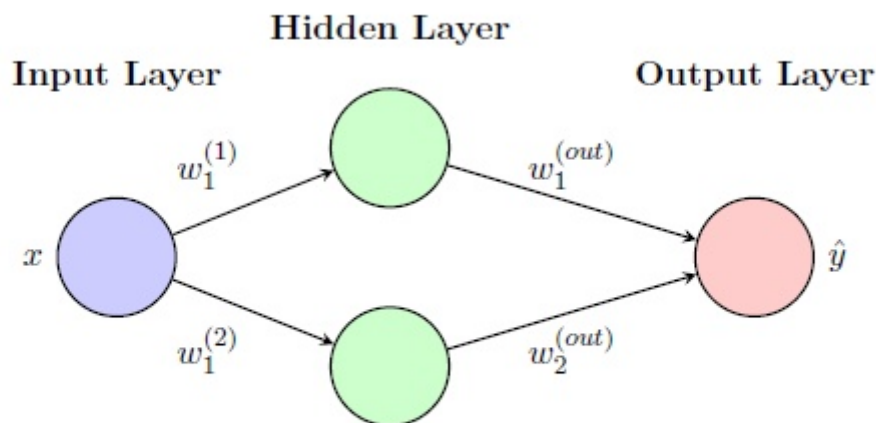
Question Label : Comprehension

Consider a fully connected neural network for a two-class (0, 1) classification task as follows:

- One input neuron x
- A single hidden layer with two neurons
- One output neuron.

The network uses the ReLU activation function in the hidden layer and sigmoid in the output layer. ReLU and the sigmoid activation is given by:

$$\text{ReLU}(x) = \max(0, x)$$
$$\sigma(x) = \frac{1}{1 + e^{-x}}.$$



Let the initial weights of the network be $w_1^{(1)} = 0.2, w_1^{(2)} = 0.1, w_1^{(out)} = 0.7, w_2^{(out)} = 1$.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 13 Question Id : 6406531852234 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

For input $x = 0.3$, what is the network output $\hat{y}$?

Options :

6406536004292. ✓ $\frac{1}{1 + e^{-0.072}}$

6406536004293. ✖ $\frac{1}{1 + e^{-0.12}}$

6406536004294. ✖ $\frac{1}{1 + e^{-0.03}}$

6406536004295. ✖ $\frac{1}{1 + e^{-0.7}}$

Question Number : 14 Question Id : 6406531852235 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

Based on $\hat{y}$, what class would the network predict for x ?

(Predict class 1 if $\hat{y} > 0.5$)

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

1

Question Id : 6406531852236 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (15 to 16)

Question Label : Comprehension

You aim to build a model that predicts whether a student will pass a course (label = 1) or fail (label = 0) by the end of the semester. Each student is described by two input features: x_1 : weekly study hours and x_2 : attendance rate. As a baseline, you train a logistic regression model that outputs a probability $p(y = 1 \mid x = (x_1, x_2))$ representing the likelihood that a student passes the course.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 15 Question Id : 6406531852237 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

How many parameters will be learned by the logistic regression model? Ignore the biases.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

2

Question Number : 16 Question Id : 6406531852238 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

You decide to train using a single hidden layer neural network with 2 hidden neurons on the same dataset and a single output neuron. How many parameters need to be learned?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

6

Question Number : 17 Question Id : 6406531852239 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

In a random forest model, let p be the total number of features and m be the number of features randomly selected at each node to determine the best split. Which of the following statements are true?

Options :

6406536004299. ✖ Increasing m reduces the correlation between any two trees in the forest.

6406536004300. ✔ Decreasing m reduces the correlation between any two trees in the forest.

6406536004301. ✔ Decreasing m can cause individual trees to underfit.

6406536004302. ✔ As m increases, the variance of the random forest model increases.

6406536004303. ✖ As m increases, the variance of the random forest model decreases.

Question Number : 18 Question Id : 6406531852240 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Consider the following one-dimensional dataset with binary labels $y \in \{+1, -1\}$.

Points	x	y
1	1	+1
2	2	+1
3	3	-1
4	4	-1
5	5	+1

AdaBoost uses decision stumps as weak learners. Assume the initial distribution of the dataset assigns equal weights to all data points, i.e., $D_0(i) = \frac{1}{5}$, for $i = 1, 2, 3, 4, 5$. At iteration $t = 1$, AdaBoost chooses the decision stump

$$h_1(x) = \begin{cases} +1, & x < 3, \\ -1, & x \geq 3. \end{cases}$$

To train the second decision stump, which pair of points will be assigned equal weights to create the training dataset?

Options :

6406536004304. ✖ Points 1 and 5

6406536004305. ✖ Points 3 and 4

6406536004306. ✔ Points 1 and 2

6406536004307. ✖ Points 2 and 5